

Part I

SUMMARY of Antiderivatives:

Summary

February 21, 2026

Summary

Suppose an object is traveling along a straight line, with displacement given by a function $s(t)$.

- The *average velocity*, v_{av} , of the object between time $t = a$ and time $t = b$ is given by

$$\begin{aligned}v_{av} &= \frac{\Delta s}{\Delta t} \\&= \frac{s(b) - s(a)}{b - a}\end{aligned}$$

- The *instantaneous velocity* of the object at time $t = a$ is given by

$$\begin{aligned}v(a) &= \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} \\&= \lim_{t \rightarrow a} \frac{s(t) - s(a)}{t - a} \\&= \lim_{t \rightarrow a} v_{av}(t)\end{aligned}$$

where $v_{av}(t)$ denotes the average velocity on the interval $[a, t]$ if $t > a$, and $[t, a]$ if $t < a$.

Part II

Recitation Questions

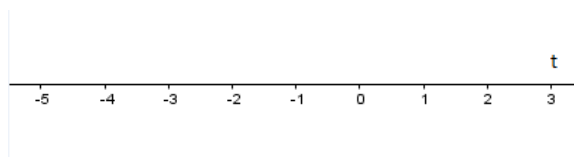
Problem 1

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Problem 1

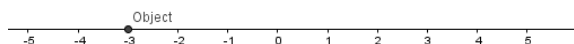
Problem 1 The position, $s(t)$, of an object moving along a horizontal line is given by $s(t) = t^2 - 4$, where s is in meters and t is in seconds, $0 \leq t < 5$.

- (a) Mark the position of the object on the line at time $t = 1$:



Explanation. $s(1) = 1^2 - 4 = -3$

Position of object on line
at time $t = 1$



- (b) Find the average velocity, v_{AV} , of the object during the time interval $[1, 3]$.

Explanation. The average velocity over $[1, 3]$ is

$$\frac{s(3) - s(1)}{3 - 1} = \frac{5 - (-3)}{2} = \frac{8}{2} = 4 \text{ m/s}$$

- (c) Compute the average velocity, $v_{AV}(t)$, of the object during the time interval

- (i) $[1, t]$, for $1 < t < 5$;

Explanation. The average velocity, $v_{AV}(t)$ over $[1, t]$ is

$$\begin{aligned} v_{AV}(t) &= \frac{s(t) - s(1)}{t - 1} = \frac{(t^2 - 4) - (-3)}{t - 1} \\ &= \frac{t^2 - 1}{t - 1} = t + 1, \quad 1 < t < 5 \end{aligned}$$

- (ii) $[t, 1]$, for $0 \leq t < 1$.

Explanation. The average velocity over $[t, 1]$ is

$$\begin{aligned} v_{AV}(t) &= \frac{s(1) - s(t)}{1 - t} = \frac{(-3) - (t^2 - 4)}{1 - t} \\ &= \frac{1 - t^2}{1 - t} = 1 + t, \quad 0 < t < 1 \end{aligned}$$

$$\text{Note: } v_{AV}(t) = \frac{s(1) - s(t)}{1 - t} = \frac{s(t) - s(1)}{t - 1} = 1 + t, \quad 0 \leq t < 5$$

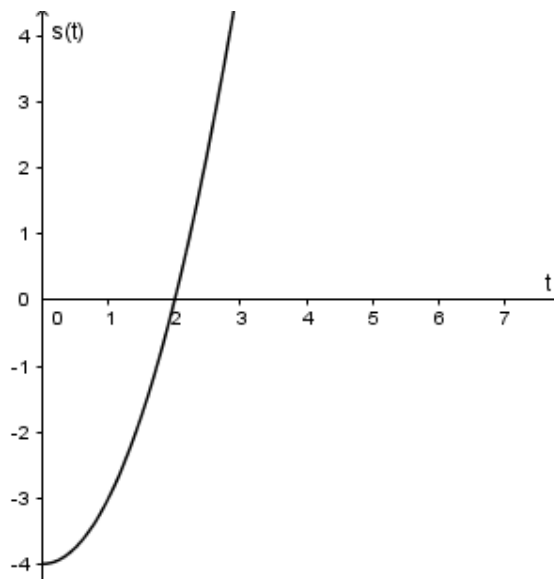
Problem 1

- (d) Find the instantaneous velocity, v_{inst} , of the object at $t = 1$. Justify your answer.

Explanation. The instantaneous velocity of the object at $t = 1$ is given by $\lim_{t \rightarrow 1} \frac{s(t) - s(1)}{t - 1}$. This limit has form $\frac{0}{0}$.

$$\begin{aligned} v_{\text{inst}} &= \lim_{t \rightarrow 1} v_{\text{AV}}(t) = \lim_{t \rightarrow 1} \frac{s(t) - s(1)}{t - 1} \\ &= \lim_{t \rightarrow 1} (t + 1) = 2. \end{aligned}$$

- (e) The position-time graph of the function s is given in the figure below.



- (i) Assume P is a point on the graph of s . Fill in the blank.

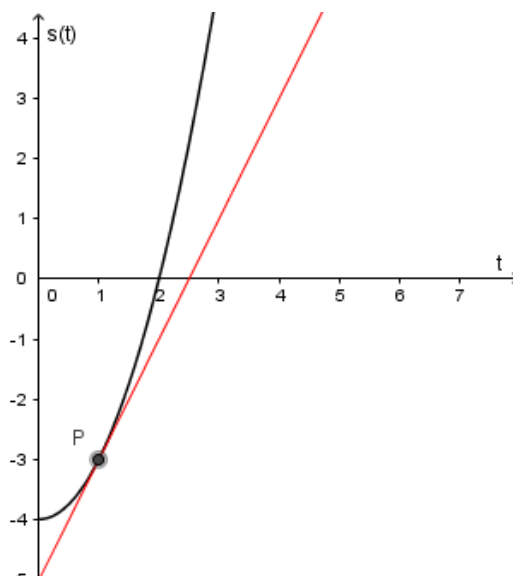
$$P = (1, \underline{\quad}).$$

Explanation. $P = (1, -3)$

- (ii) Plot the point P and draw the tangent line at this point in the figure above.

Explanation.

Problem 1



- (iii) Find the slope, m_{tan} , of the tangent line in part (ii). Explain.

Explanation. The slope of the tangent line at $t = 1$ is the same as the instantaneous velocity at $t = 1$. Therefore $m_{\text{tan}} = v_{\text{inst}} = 2$.

Problem 2

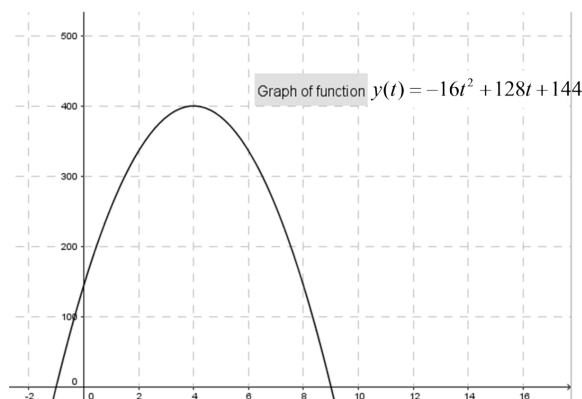
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Problem 2

Problem 2 Part of the given parabola can be used to model the “position-time” graph of a ball thrown straight up into the air. The graph gives the height of the ball in feet t seconds after being thrown into the air.

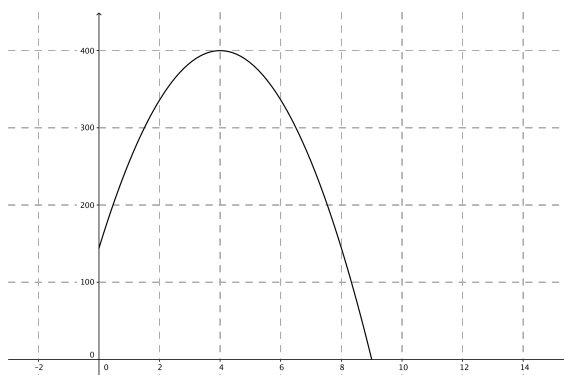
Let the function f be defined by $f(t) = -16t^2 + 128t + 144$.

Use this graph, and the given function, f , to answer the following questions.



- (a) Mark the part of the parabola that can be used to model the position of the ball.

Explanation.



- (b) What are the units on the t axis? What are the units on the y axis?

Explanation. The units on the t axis are “seconds” (for time), while the units on the y axis are “feet” (for height).

- (c) If you were watching a movie of the ball being thrown, is the graph a picture of the path that the ball follows? Why or why not?

Explanation. No, the position-time graph is *not* the path the ball follows. The graph shows the height of the ball at a given *time*. The ball is thrown

Problem 2

straight up and has no horizontal movement so its path is on a vertical line.

- (d) Let $f(t)$ denote the height of the ball at time $t, t \geq 0$. What is the height of the ball at time $t = 0$?

Explanation. The height can be found by finding $f(0)$

$$f(0) = -16(0)^2 + 128(0) + 144$$

$$f(0) = 144 \text{ feet}$$

- (e) When will the ball hit the ground?

Explanation. The ball will hit the ground when the height $f(t)$ equals zero.

$$0 = -16t^2 + 128t + 144$$

$$0 = -16(t^2 - 8t - 9)$$

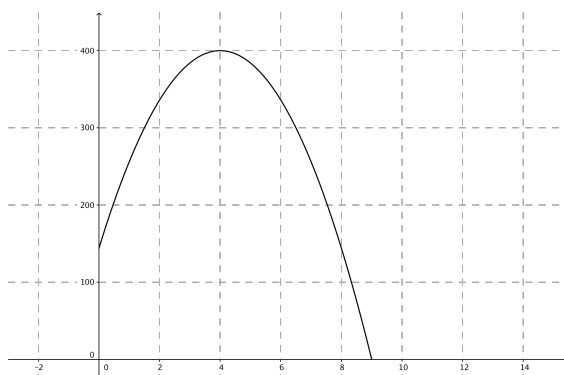
$$0 = -16(t + 1)(t - 9)$$

$$t = -1, 9$$

The ball will hit the ground at $t = 9$ or 9 seconds after the ball is thrown into the air.

- (f) What is the domain of the position function, f , of the ball?

Explanation. The domain of f is the interval $[0, 9]$. The ball is thrown straight up at $t = 0$ and hits the ground at $t = 9$. With this domain, the position-time graph of the ball is given by



- (g) Use the table of values to find the average velocity of the ball between

Problem 2

$t = 8.9$ and $t = 9$ seconds.

t	$\approx f(t)$
8.9	15.84
8.99	1.6
8.999	0.159984
8.9999	0.015998
9	0

Explanation. The average velocity of the ball between $t = 8.9$ seconds and $t = 9$ seconds is

$$\frac{f(9) - f(8.9)}{9 - 8.9} = \frac{0 - 15.84}{0.1} = -158.4 \text{ feet per second}$$

- (h) Use the table of average velocities to approximate the instantaneous velocity of the ball when it hits the ground.

Time Interval	Average Velocity
$[8.9, 9]$	$\frac{f(9) - f(8.9)}{.1} = \frac{0 - 15.84}{.1} = -158.4$
$[8.99, 9]$	$\frac{f(9) - f(8.99)}{.01} = \frac{0 - 1.5984}{.01} = -159.84$
$[8.999, 9]$	$\frac{f(9) - f(8.999)}{.001} = \frac{0 - .159984}{.001} = -159.984$
$[8.9999, 9]$	$\frac{f(9) - f(8.9999)}{.0001} = \frac{0 - .0159998}{.0001} = -159.998$

Explanation. The instantaneous velocity of the ball hitting the ground appears to be -160 ft/sec.

- (i) Compute $v_{AV(t)}$ the average velocity of the ball on the time interval $[t, 9]$, where $t < 9$.

Explanation.

$$\begin{aligned}
 v_{AV(t)} &= \frac{f(9) - f(t)}{9 - t} \\
 &= \frac{0 - (-16t^2 + 128t + 144)}{9 - t} \\
 &= \frac{16t^2 - 128t - 144}{9 - t} \\
 &= \frac{16(t^2 - 8t - 9)}{9 - t} \\
 &= \frac{16(t - 9)(t + 1)}{9 - t} \\
 &= \frac{-16(9 - t)(t + 1)}{9 - t} \\
 &= -16(t + 1) \text{ feet per second}
 \end{aligned}$$

(j) Compute $v(9)$, the **instantaneous** velocity of the ball at $t = 9$.

Explanation.

$$\begin{aligned}
 v(9) &= \lim_{t \rightarrow 9^-} v_{AV(t)} \\
 &= \lim_{t \rightarrow 9^-} \frac{f(9) - f(t)}{9 - t}
 \end{aligned}$$

Notice that this is an indeterminate form, with form $\frac{0}{0}$.

$$\begin{aligned}
 v(9) &= \lim_{t \rightarrow 9^-} v_{AV(t)} \\
 &= \lim_{t \rightarrow 9^-} \frac{f(9) - f(t)}{9 - t} \\
 &= \lim_{t \rightarrow 9^-} -16(t + 1) \\
 &= -16(9 + 1) \\
 &= -160 \text{ feet per second}
 \end{aligned}$$